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CSC 6302 - Database Principles

Week 5 - Amanda Menier

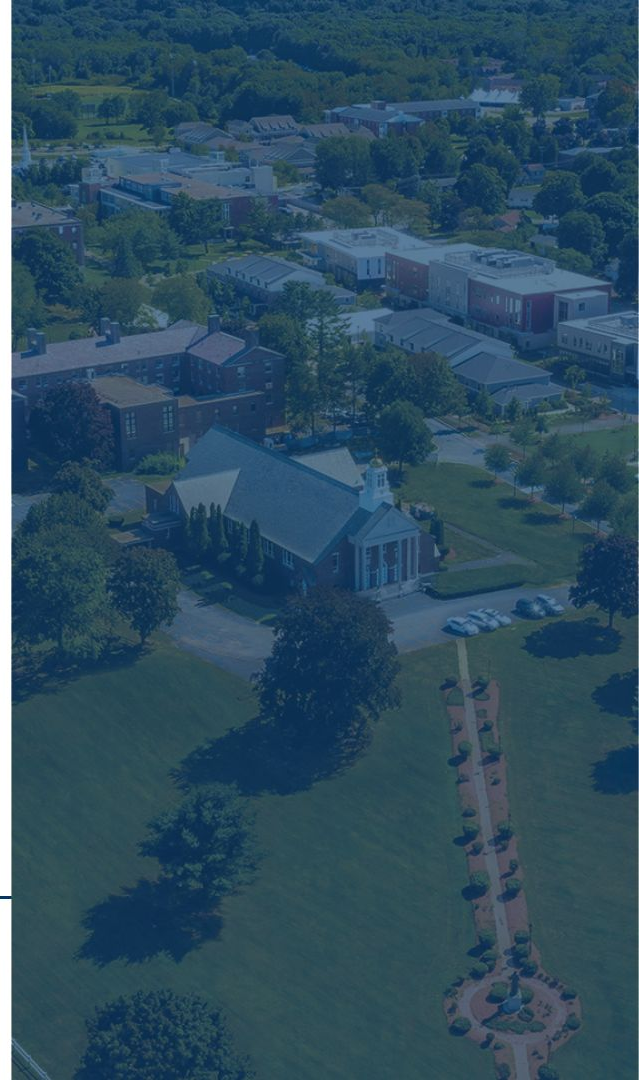
Agenda

Discussion Topics:

- Next few weeks
- MVC review
- A new analogy
- Common view packages
- Widgets
- View examples
- Project 5 instructions
- Project 4 and earlier review



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Future Projects

This week: Adding View Layer
Next week: Final project written plan
Week 7: Final project DB
Week 8: Final project GUI application



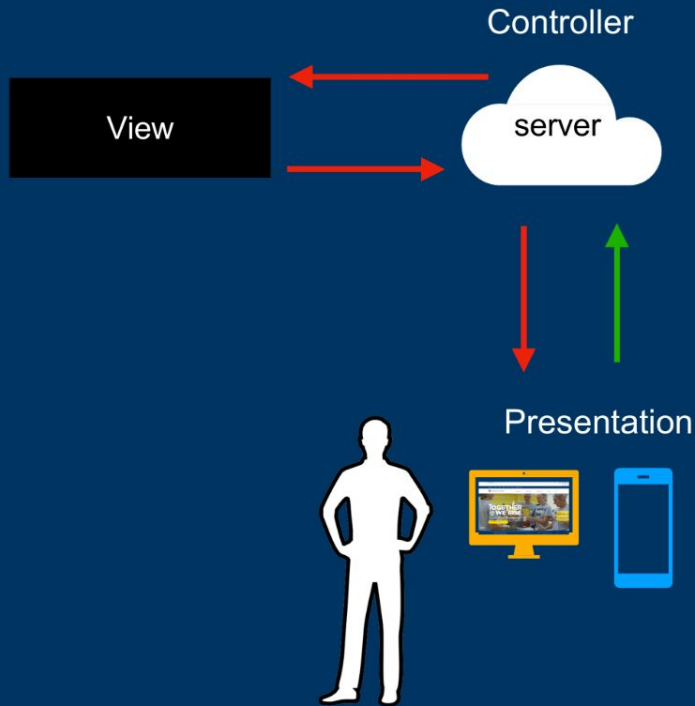
Project 4 Feedback

- 1) Config & connections
- 2) Too much copying of the sample program
- 3) The new vessel/passenger combo wasn't added to the database. They would need to be added separately first, or you would need code in your add trips function to add them if they don't exist.



MVC

Model - View - Controller



Key Responsibilities

Data Presentation: Formatting and displaying data retrieved from the model layer in a way that is understandable and useful to users.

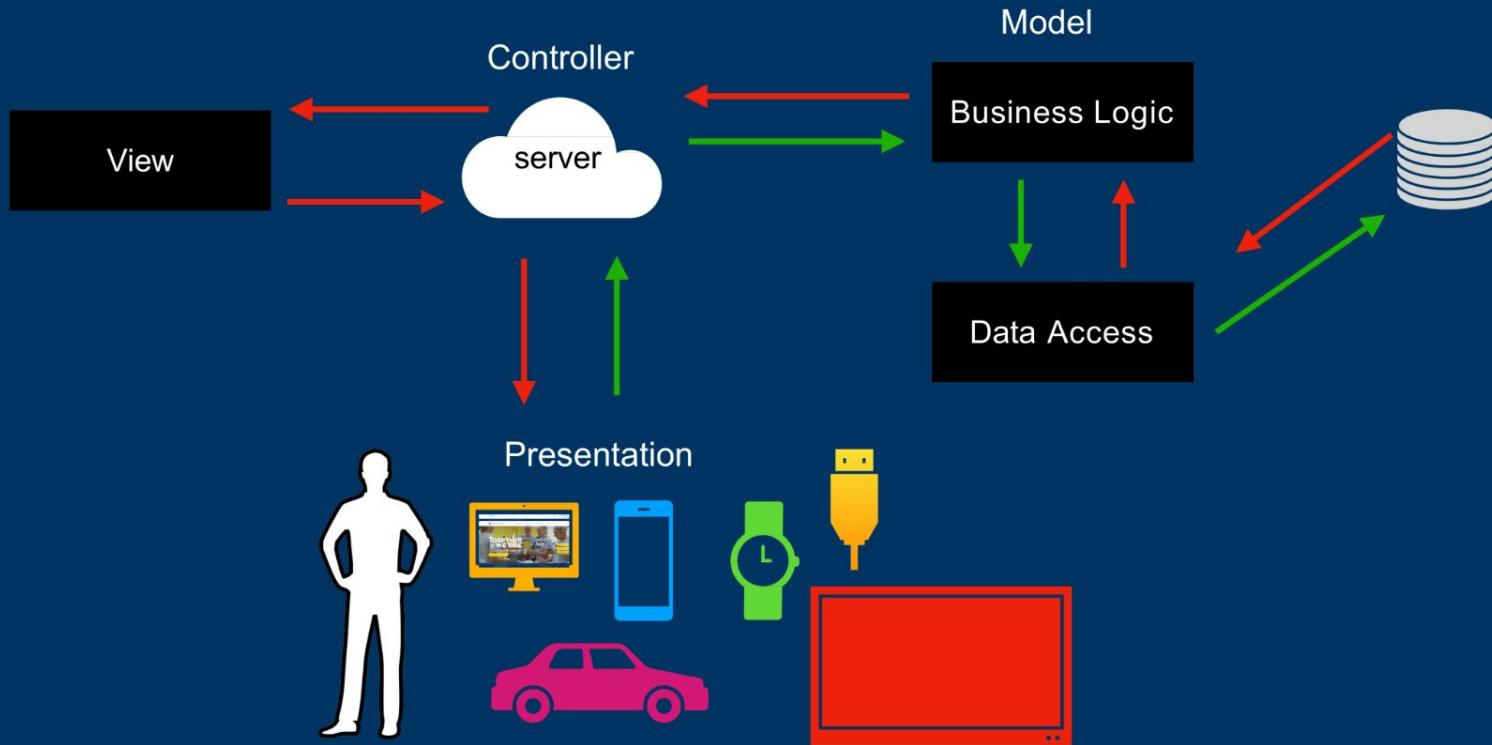
User Interaction: Providing user interface components (like buttons, forms, and menus) that allow users to interact with the application.

Input Validation: Ensuring that user input is valid and providing immediate feedback on errors or issues.

User Experience Design: Creating a responsive and intuitive interface that enhances user engagement and satisfaction.

MVC

Model - View - Controller



Department Store Analogy

- View layer
 - Department shelves
 - Customers pick from the available merchandise
 - Goods are ready to buy, easy to see, attractively displayed
- Business Logic Layer
 - Manager making decisions about to buy, when, and where to put it in the store
 - Manager putting in orders to the shipping/receiving department based on customer requests
 - Sometimes these are frequent predefined orders, sometimes they are special orders
- Data Access Layer
 - Shipping/receiving department
 - Takes the orders from the manager and translates them into purchase orders for the supplier
 - When the order comes back, takes items off truck, unpacks them, removes plastic
 - Stages the items for the Manager who then directs them to the proper shelves



Common Python GUIs

- Tkinter:
 - Standard GUI toolkit for Python and comes bundled with most Python installations.
- PyQt
 - A set of Python bindings for the Qt application framework. It is powerful and allows developers to create cross-platform applications with rich graphical interfaces.
- Toga
 - An open-source, cross-platform GUI toolkit for Python, developed as part of the BeeWare project. It allows developers to create native applications for Windows, macOS, Linux, iOS, and Android using a single codebase. Toga aims to provide a simple and consistent API across all platforms.
- Flask
 - A lightweight and flexible web framework for Python, designed to make it easy to build web applications and APIs (Serve HTML, JS, and CSS)
- PySimpleGUI
 - Uses lists to organize layouts - good if you are having trouble with the widget-heavy models



Widgets

Widgets are the building blocks of many GUI builders.
Think of them as a nest or tree of content

Main Window

```
|----- Frame 1
      |----- Button 1
      |----- Button 2
|----- Frame 2
      |----- Entry Box 1
      |----- Entry Box 2
|----- Frame 3
      |----- Results Display Area
```



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Sample Function/Layer

Goal – Get all students taught by George Feeny

Hierarchy

Main function – Start the program

View – ## Create interface, including text boxes to enter teacher name

```
window = tk.Tk()
fnamebox = tk.Entry...; fname_value = "George"
lnamebox = tk.Entry...; lname_value = "Feeny"
## define button to get information from text boxes and do business logic with it
btn = tk.Button(window, text = "Submit", command = lambda:
    BLL.getStudents(fname_value.get(), lname_value.get()))
## define method to display results
def showResults(results):
    for x in results:
        tk.Label(window, x)
```



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Sample Function/Layer

Goal – Get all students taught by George Feeny

Hierarchy

BLL – ## Create function that takes information from view and packages it to send to DAL

```
def getStudents(fname, lname):  
    View.message(f"Searching for {fname} {lname}")  
    View.showResults(DAL.getDBStudents(fname, lname))
```

DAL – ## Define Function to get students

```
def getDBStudents (fname, lname):  
    cursor.execute(f"CALL getStudents(getTeacherId({fname},{lastname}))")
```

Database – ## Create & call Functions & Stored Procedures

```
CREATE getTeacherId(fname VARCHAR(20), name VARCHAR(20));  
CREATE getStudents(teacherId INT);  
CALL getStudents(getTeacherId("George", "Feeny");
```



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Let's Look at Some Example Programs



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Front End Example: Chewy.com

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This promo code (missyou15) was already redeemed on a previous order.

Project 5

It's time to add a view layer that lets us add passengers, vessels, and trips.

